# Unit ID: 679b18cdaaf593b2823f3546

# Unit 2 - Concept of Length

## PDF Documents

### PDF Document 1: 1738217675394-Chapter 4 - Unit 2 - Concept of Length.pdf

This PDF document is included in the unit materials.

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## UNIT 2 CONCEPT OF LENGTH

The concept of length refers to measuring the distance from one point to another. Length helps compare or describe the size or distance of objects, and it plays an important role in daily life, as well as in science and engineering.

**The main units of length are as follows:**

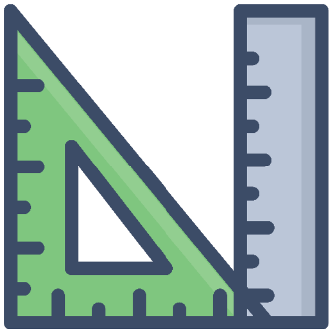
| MAIN UNITS OF LENGTH MILLIMETER (MM) | UNIT CONVERSIONS |
|---|---|
| 1. Centimeter (cm) | • 1 inch ≈ 2.54 cm |
| 2. Meter (m) | • 10mm = 1cm |
| 3. Kilometer (km) | • 100cm = 1m |
| 4. Inch (inch) | • 1,000m = 1km |

We often encounter situations in everyday life where we need to measure length. For example, we may need to measure our height at an amusement park to ride certain attractions. Measuring height is one way of measuring length, but there are different methods.

## DEVICES FOR MEASURING LENGTH

### RULER

A ruler is the most basic length-measuring tool for measuring straight distances. It has centimeter (cm) and millimeter (mm) markings, allowing for precise measurement of short distances or small objects.

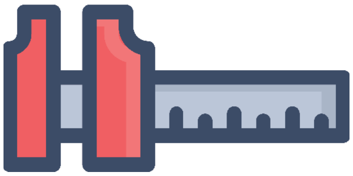


### MEASURING TAPE

A flexible tool for measuring length, measuring tapes are useful for long distances or objects with curves. They typically show centimeter and inch units and are widely used in settings ranging from households to construction sites.

### VERNIER CALIPER

A vernier caliper is a precise tool that can measure down to 0.1 mm or even 0.01 mm. It measures not only the distance between two points but also the inner and outer diameters and depth, making it valuable in various industrial fields.



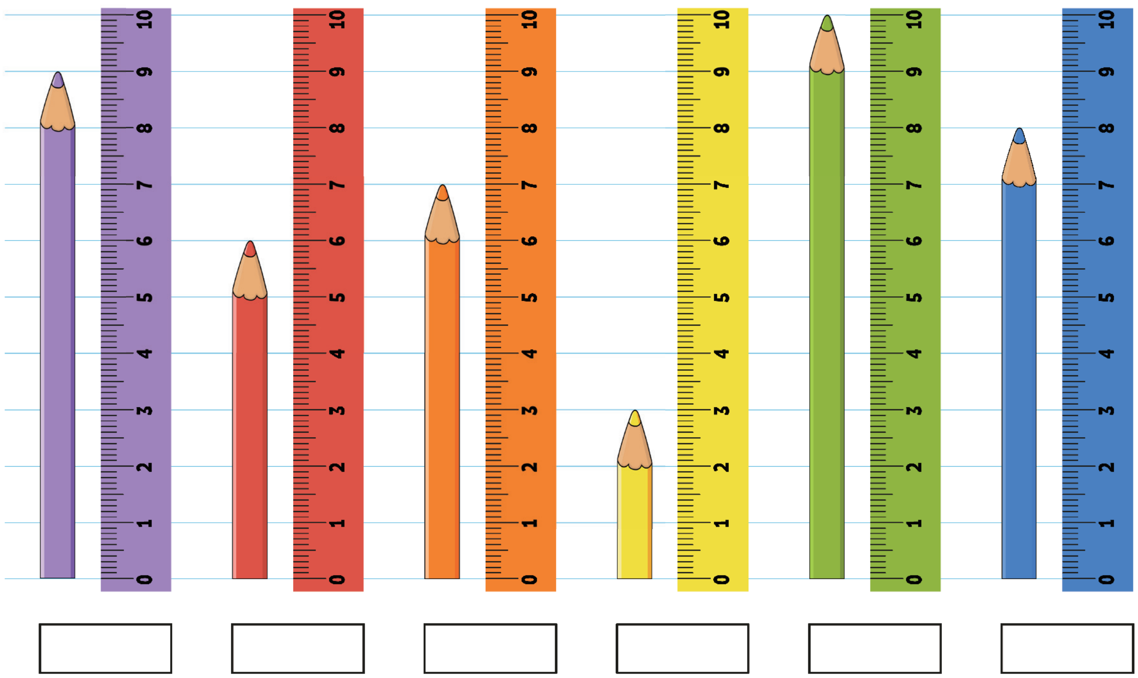
### TIME-OF-FLIGHT SENSOR

The ToF sensor calculates distance by measuring the time light hits an object and bounces back. Commonly used in electronic devices, ToF sensors can measure distances without contact and are widely used in fields like robotics, smartphones, and autonomous vehicles.

## Activity: Let's measure length and convert units.

1 Fill in the blank spaces in the picture below. The units are in cm.

2 Convert the length of the green pencil into inches.



3 What is the giraffe's height in cm in the picture?

4 Convert the giraffe's height into meters.

5 Convert the giraffe's height into inches.